Final Course Project Instructions

Due date: December 6, 2026

1. Final Course Project Overview

Select a healthcare-related dataset from an online source. Using Python and the methods covered in the course, you will prepare, explore, and analyze the data and apply an appropriate AI or machine-learning method.

Required submissions (all due December 6, 2026):

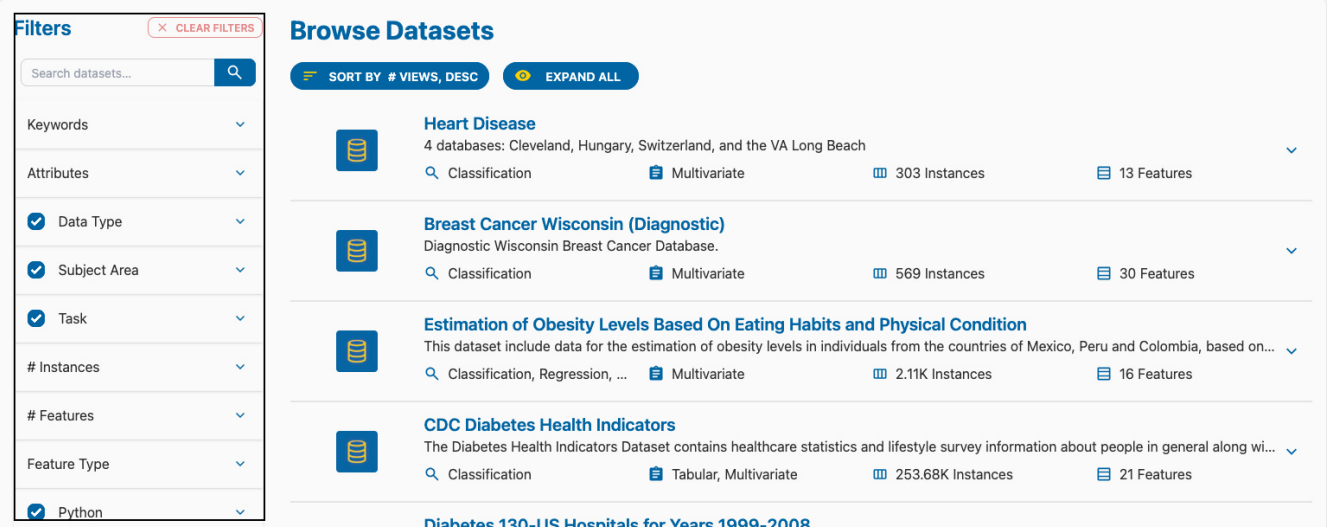
- **PowerPoint presentation:** Summarize your project and findings.
- **Microsoft Word report:** Describe the dataset, methods, analysis, and results.
- **Jupyter Notebook (`.ipynb`):** Include your Python code and analysis.

2. Finding a Healthcare Dataset

The **UCI Machine Learning Repository** provides datasets that can be used for the course project. Use the filters to find healthcare data that are easy to access and practical to analyze in Python.

Step 1. Open the repository

Go to https://archive.ics.uci.edu/datasets/. The **Filters** panel appears on the left, and the available datasets appear on the right.

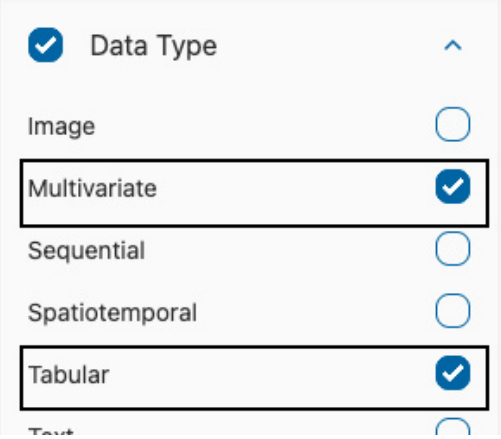


Step 2. Apply the recommended filters

Use the filter settings on the following page to narrow the results to manageable healthcare datasets suitable for a beginner-level project.

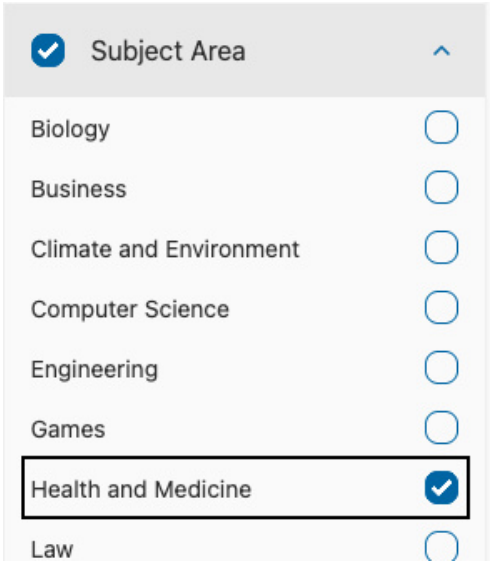
Recommended Dataset Filters

Data Type: Multivariate and Tabular



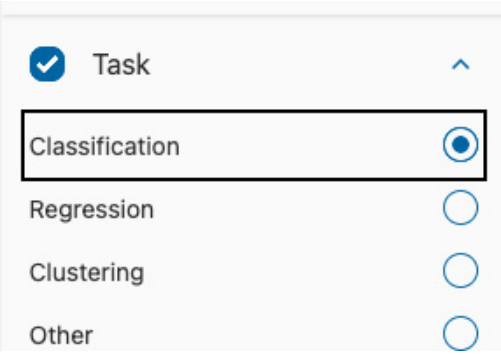
Choose these to find datasets with **multiple variables** arranged in **rows and columns**, which are easier to work with using Pandas.

Subject Area: Health and Medicine



Limits the results to **healthcare-related topics** suitable for the course project.

Task: Classification



Recommended when predicting a **category**, such as Yes/No. You may choose **Regression** instead if your outcome is numerical.

Python: Available for Python Import



Finds datasets with Python import instructions, making it easier to **load the data directly into Jupyter Notebook**.

Step 3. Check the dataset size

Use **# Instances** (rows) and **# Features** (input columns) to judge the size of the dataset. For a first project, a dataset with a few hundred to a few thousand rows and a moderate number of columns is generally easier to inspect and analyze.

Step 4. Review the results

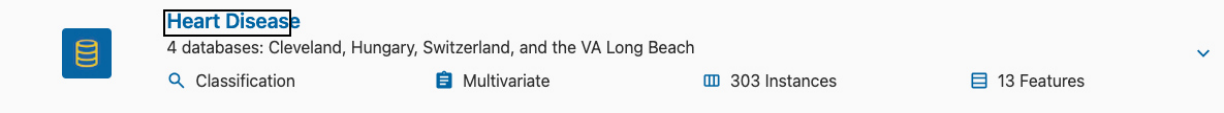
Click a dataset title to open its page. Before selecting it, check the description, number of records, variables, outcome (target), and missing-value information.

3. Example: Heart Disease

The **Heart Disease** dataset appears first in the filtered results shown in the screenshots. Use it as an example of how to open a dataset, read its information, and locate the Python import code.

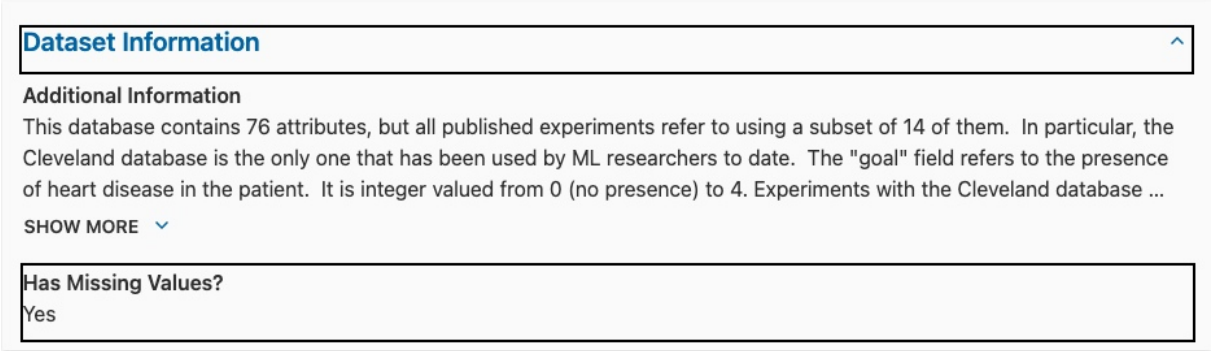
1. Open the dataset

Click **Heart Disease**, the first dataset listed in the results.



2. Review the dataset information

The Heart Disease dataset page lists **303 records** and **13 features**. Read the **Dataset Information** section to understand the dataset and check for missing values. Also review the **Variables Table** to see what each variable represents.



3. Find the Python import code

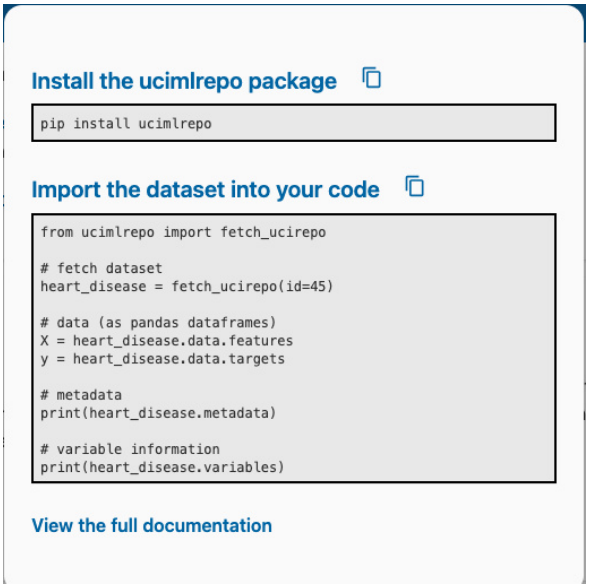
Click **IMPORT IN PYTHON** on the right side of the dataset page. A window will open with the commands to copy into Jupyter Notebook.



4. Loading the Dataset into Jupyter Notebook

Step 1. Copy the import instructions

The **IMPORT IN PYTHON** window contains a package-installation command and the Python code for accessing the dataset.



Copy these two parts:

1. The command beneath **Install the ucimlrepo package**.
2. The code beneath **Import the dataset into your code**.

For this example, the Heart Disease dataset ID is `45`.

Step 2. Install the package

In a Jupyter Notebook code cell, enter the command and press **Shift + Enter**:

```
!pip install ucimlrepo
```

```
[1]: !pip install ucimlrepo
     Requirement already satisfied: ucimlrepo in /opt/anaconda3/lib/python3.14/site-packages (0.0.7)
     Requirement already satisfied: pandas>=1.0.0 in /opt/anaconda3/lib/python3.14/site-packages (from ucimlrepo) (3.0.3)
     Requirement already satisfied: certifi>=2020.12.5 in /opt/anaconda3/lib/python3.14/site-packages (from ucimlrepo) (2026.7.22)
```

If the output says "Requirement already satisfied," the package is already installed.

Step 3. Load and preview the data

In a new code cell, run the following code. `X` contains the input variables and `y` contains the target variable.

```
from ucimlrepo import fetch_ucirepo

heart_disease = fetch_ucirepo(id=45)
X = heart_disease.data.features
y = heart_disease.data.targets

X.head()          # first 5 rows of input variables
```

Run `y.head()` in another cell to view the target values.

5. What Students Should Do Next

Review the variable definitions, check the number of rows and columns, identify missing values, explore the data, and apply an appropriate course method. Include your dataset source and citation in the report.

Dataset shown in this guide: Heart Disease, UCI Machine Learning Repository.